

Homework (Carolina Reaper)

- Q1.** A rhythm pattern has 3 repeats in 12 beats. What is the rate of repeats per beat?
(A) $\frac{1}{4}$
(B) $\frac{1}{3}$
(C) $\frac{1}{2}$
(D) $\frac{1}{6}$
(E) $\frac{1}{12}$
- Q2.** If a painter can finish a 16 ft by 20 ft canvas in 8 hours, how many hours to finish a 12 ft by 15 ft canvas at the same rate?
(A) 3 hours
(B) 4.5 hours
(C) 6 hours
(D) 7.2 hours
(E) 9 hours
- Q3.** The frequency of a sound wave is 250 Hz. How many cycles per second is this sound wave?
(A) 100 cycles
(B) 125 cycles
(C) 200 cycles
(D) 250 cycles
(E) 500 cycles
- Q4.** A music school has 18 teachers. If the student-to-teacher ratio is 12 : 1, how many students are there in total?
(A) 108 students
(B) 144 students
(C) 180 students
(D) 216 students
(E) 288 students
- Q5.** The length of a song is 4 minutes and 15 seconds. If this song is played 3 times in a row, what is the total time in seconds?
(A) 765 seconds
(B) 780 seconds
(C) 795 seconds
(D) 810 seconds
(E) 825 seconds
- Q6.** An art piece has a width that is 2 times its height. If the width is 12 ft, what is the area of the art piece in square feet?
(A) 24 sq ft
(B) 36 sq ft
(C) 48 sq ft
(D) 72 sq ft
(E) 96 sq ft
- Q7.** If x is inversely proportional to y , and $y = 4$ when $x = 3$, what is the value of y when $x = 2$?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 12
- Q8.** A painter works at a rate of 2 walls per 5 hours. How many hours will it take to paint 12 walls at this rate?
(A) 15 hours
(B) 20 hours
(C) 25 hours
(D) 30 hours
(E) 35 hours

Open Ended Questions (FRQ)

- Q9.** A sound wave has a frequency of x Hz. If the time period is 2 seconds, find the value of x and explain the relationship between frequency and time period.
- Q10.** The cost of recording a song is
- 200
- plus
- 50
- per hour of studio time. If x is the number of hours of studio time, write an equation to represent the total cost and graph the equation with x on the horizontal axis and cost on the vertical axis.

Chef's Tips

- Please ensure students have the necessary calculators and tools for the test.
- Remind students that they must show all work for free response questions.
- Emphasize that all answers should be clearly labeled and easy to read.